Day 065 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Explain Balance and Inverse Operations

Standards: 6.EE.B.7

Learning target: I can explain Balance and Inverse Operations, and explain and check my reasoning.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Subtract p from both sides to preserve equality. Then substitute the answer into the original equation. The value of the left side must equal the right side. Add p to both sides to undo subtraction. Check by subtracting p from your answer. Values in this lesson are nonnegative rational numbers. Divide both sides by the nonzero coefficient p. Keep the equation balanced. Check by multiplying the proposed solution by p.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 11

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01 / Learn

Subtract p from both sides to preserve equality. Then substitute the answer into the original equation. The value of the left side must equal the right side. Add p to both sides to undo subtraction. Check by subtracting p from your answer. Values in this lesson are nonnegative rational numbers. Divide both sides by the nonzero coefficient p. Keep the equation balanced. Check by multiplying the proposed solution by p.

Worked example 1

Solve x + 7 = 14. Check in the original equation.

1. Subtract p from both sides to preserve equality.

2. Then substitute the answer into the original equation.

3. Subtract 7 on both sides.

4. Then x = 7

5. substitution gives 14 on each side.

6. State the result in the form requested: x = 7.

Answer: x = 7

Worked example 2

Solve 3/2 × x = 15/4. Select every representation of the solution.
A. 7/2
B. 5/2
C. 10/4
D. 9/2

1. Divide both sides by the nonzero coefficient p.

2. Use the inverse operation on both sides.

3. The solution is 5/2

4. the doubled numerator and denominator name the same value.

5. Substitution verifies the solution.

Answer: B, C - 5/2; 10/4

Day 065 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Solve x + 5 = 7. Check in the original equation.

02 / Constructed response

Solve x - 3 = 2. Check in the original equation.

03 / Constructed response

Solve 6 × x = 18. Check in the original equation.

04 / Constructed response

Solve x + 8 = 19/2. Check in the original equation.

05 / Constructed response

Solve x - 6 = 3/2. Check in the original equation.

06 / Constructed response

Solve 8 × x = 8. Check in the original equation.

07 / Constructed response

Solve x + 4 = 11/2. Check in the original equation.

08 / Multiple select

Solve x - 5/2 = 1. Select every representation of the solution.
A. 9/2
B. 7/2
C. 11/2
D. 14/4

09 / Multiple select

Solve 2 × x = 9. Select every representation of the solution.
A. 9/2
B. 13/2
C. 11/2
D. 18/4

10 / Constructed response

Solve x + 1 = 9/2. Check in the original equation.

Day 065 | Unit 05 | Answer key

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1. x = 2

Subtract 5 on both sides. Then x = 2; substitution gives 7 on each side.

6.EE.B.7 | Constructed response

2. x = 5

Add 3 on both sides. Then x = 5; substitution gives 2 on each side.

6.EE.B.7 | Constructed response

3. x = 3

Divide by 6 on both sides. Then x = 3; substitution gives 18 on each side.

6.EE.B.7 | Constructed response

4. x = 3/2

Subtract 8 on both sides. Then x = 3/2; substitution gives 19/2 on each side.

6.EE.B.7 | Constructed response

5. x = 15/2

Add 6 on both sides. Then x = 15/2; substitution gives 3/2 on each side.

6.EE.B.7 | Constructed response

6. x = 1

Divide by 8 on both sides. Then x = 1; substitution gives 8 on each side.

6.EE.B.7 | Constructed response

7. x = 3/2

Subtract 4 on both sides. Then x = 3/2; substitution gives 11/2 on each side.

6.EE.B.7 | Constructed response

8. B, D - 7/2; 14/4

Use the inverse operation on both sides. The solution is 7/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

9. A, D - 9/2; 18/4

Use the inverse operation on both sides. The solution is 9/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

10. x = 7/2

Subtract 1 on both sides. Then x = 7/2; substitution gives 9/2 on each side.

6.EE.B.7 | Constructed response